

FLUIDBLCX // SOVEREIGN WEB3 EXECUTION ARCHITECTURE

CONSENSUS PROTOCOL
MYSTICETI L1 (11ms)

CIPHER & SHARDING
AES-256-GCM • RS(8,6)

MASTER TECHNOLOGY, ARCHITECTURE & LEXICON SPECIFICATION — 35 CORE SUBSYSTEMS

SPEC VERSION: v3.4.0 (2026 ARCH)

NETWORK: SUI MAINNET & TESTNET

#	Category	Technology / Concept	Target Module	Technical Definition & Architectural Role	Technical Specifications
01	Blockchain Core	Sui L1 Network	Core Infrastructure	High-throughput Layer 1 blockchain optimized for fast execution horizontal scalability and low latency object storage.	Move VM • parallel transaction processing • sub-second finality
02	Consensus	Mysticeti Consensus Engine	L1 Consensus Layer	Next-generation DAG-based consensus protocol powering Sui reducing validator commit latency to sub-second speeds.	11ms - 390ms latency • Byzantine Fault Tolerant (BFT) • asynchronous DAG rounds
03	Smart Contracts	Sui Move	Compiler & Smart Contracts	Object-centric smart contract programming language focused on safe resource management asset ownership and bytecode verification.	Linear logic • object-centric model • capability security • byte-level verification
04	Execution Model	Programmable Transaction Blocks (PTB)	Compiler & DeepBook Trading	Atomic multi-instruction execution bundles allowing developers to chain up to 1024 commands in a single transaction with output passing.	Atomic execution • zero gas wastage on chained calls • composable commands
05	Data Model	Object-Centric Storage Model	Sui Core State	Data architecture where everything is an object with globally unique 32-byte UID versioning and explicit ownership (Owned vs Shared).	UID • TxContext • shared objects • dynamic field extensions
06	Cryptography	ECDSA SEC_P256	Gateway & Security	Elliptic Curve Digital Signature Algorithm over the NIST P-256 prime curve used for high-security transaction authorization.	256-bit key length • asymmetric keypairs • cryptographic signature verification
07	Cryptography	Ed25519	Wallet & Key Management	High-speed Edwards-curve digital signature algorithm standard for fast Web3 keypair creation and transaction signing.	128-bit security level • resistant to side-channel attacks • deterministic
08	Storage & Encryption	Walrus Protocol	Walrus Decentralized Vault	Decentralized blob archival protocol designed for large binary payloads on Sui using erasure coding and certified epochs.	Epoch-based storage • high availability • decentralized shard routing
09	Cryptography	AES-256-GCM	Walrus Backend Storage	Advanced Encryption Standard in Galois/Counter Mode providing authenticated symmetric encryption and cryptographic integrity verification.	256-bit symmetric cipher • 96-bit initialization vector (IV) • 128-bit authentication tag
10	Coding Theory	Reed-Solomon (8 6) Erasure Coding	Walrus Shard Engine	Algebraic error-correction algorithm that partitions payloads into 6 primary data shards + 2 parity shards tolerating loss of any 2 shards.	N=8 K=6 P=2 • deterministic matrix multiplication • fault recovery
11	Data Structures	Merkle Tree & Merkle Root	Walrus Proof Engine	Cryptographic binary hash tree where leaf nodes store SHA-256 shard digests and parent nodes hash combined pairs up to a root hash.	SHA-256 hashing • logarithmic inclusion proofs $O(\log N)$ • tamper-proof integrity
12	DeFi & Liquidity	DeepBook v3 CLOB	DeepBook Trading Engine	On-chain Central Limit Order Book with ultra-low latency matching maker rebates and native Move atomic execution.	Zero orderbook indexing latency • order ID hashing • depth ladders • maker-taker fees
13	3D Graphics	Geodesic Mysticeti Sphere	Institutional Gateway	Real-time 3D polyhedral visualization of the consensus state featuring glowing icosahedral faces wireframe cage and orbital particles.	Three.js WebGL • IcosahedronGeometry • dynamic emissive shader shifting
14	3D Graphics	Holographic Tesseract Vault Cube	Walrus 3D Stage	Multi-tiered 3D storage vault model that physically explodes into an 8-shard constellation with dynamic laser links upon detonation.	BoxGeometry • TorusGeometry • gyroscopic rotation • dynamic laser beam vectors
15	Institutional DeFi	Multi-Wallet CSV Batch Provisioning	DeepBook Gateway	Ingestion pipeline that parses institutional .csv portfolios and assigns allocations across 10 parallel node clusters.	portfolio.csv • parallel node streaming ([NODE-SYNC-01..10]) • batch verification
16	Market Data	12-Asset L2 Matrix	CLOB Terminal Marquee	Real-time streaming market data ticker tracking spot prices 24h percentage changes and spreads for 12 major assets.	SUI • BTC • ETH • SOL • PAXG (Gold) • CETUS • NAVX • DEEP • WAL • USDC • USDT • BLCX
17	Data Visualization	HTML5 Canvas Candlestick Engine	CLOB Charting Pane	Custom high-performance 2D canvas charting engine rendering real-time OHLCV (Open High Low Close Volume) bars and price lines.	Dynamic scale normalizer • crosshair coordinates • 1m/5m/15m/1h/1D timeframe intervals

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18	● Trading Interface	Level 2 (L2) Order Book	CLOB Depth Ladder	Live visual orderbook displaying active Bid and Ask limits with proportional depth bars and real-time spread calculations.	Real-time spread-meter • ask depth (red) • bid depth (green) • size aggregation
19	● Artificial Intelligence	Autonomous AI Alpha Execution Agent	AI Alpha Pane	Machine learning-inspired market-making agent monitoring orderbook micro-slippage and generating high-confidence arbitrage signals.	94.2% Alpha confidence metric • +1.42% arbitrage spread detection • autonomous dispatch
20	● DeFi Infrastructure	Deep Liquidity Reserve Pools	CLOB Reserve Pane	Real-time automated market liquidity tracking showing Total Value Locked (TVL) pool reserve balances and base/quote ratios.	\$248.6M TVL • 14.8% APY • dual-asset liquidity reserve ratio track
21	● Audio Synthesis	Web Audio API Synthesizer	Interactive Audio Layer	Procedural client-side audio oscillator generating tactile sound effects for detonations shatters orders and cryptographic validations.	AudioContext • createOscillator • frequency modulation (sawtooth square sine triangle)
22	● Animation Engine	GSAP (GreenSock Animation Platform)	UI Interaction & Shatter	Physics-based animation library driving UI micro-interactions elastic camera shakes and file-preview shatter particle explosions.	gsap.to() • gsap.fromTo() • cubic-bezier curves • back-out springs
23	● Security & Auditing	Security Sentinel Dashboard	Threat Defense Center	Real-time monitoring hub aggregating firewall status threat intelligence metrics and smart contract vulnerability analysis.	Live network anomaly scanning • gas exhaustion guards • risk scoring engine
24	● AI Defense	Poison AI & Adversarial Prompt Firewall	Security Sentinel	Semantic filtering engine detecting prompt injection model jailbreaks and adversarial input vectors attacking LLM agents.	Token anomaly detection • prompt sanitization • behavioral boundaries
25	● Smart Contract Security	Trojan Bots & Malicious Bytecode Detector	Security Sentinel	Static analysis engine inspecting Move packages for reentrancy vectors unconstrained capability leaks and flash loan drain traps.	Move bytecode disassembler • call-graph analysis • capability privilege audit
26	● AI NLP Parsing	IntentParser Engine	Move Compiler Workspace	Natural language compiler converting conversational human financial intents into valid structured Move PTB commands.	Lexical parsing • token classification • multi-step transaction command synthesis
27	● Visual Programming	Dynamic SVG Graph Connectors	Move Compiler Workspace	Interactive visual programming workspace connecting Inputs Guardrails Commands and Outputs via dynamic cubic Bezier curves.	SVG • cubic bezier calculation M x1 y1 C cx1 cy1 cx2 cy2 x2 y2 • auto-redraw
28	● Asset Management	QuantX Portfolio & Algorithmic Trading	Wallet & Trade Terminal	Institutional portfolio tracker and algorithmic trading window with automated rebalancing and risk-weighted allocation.	Multi-token balance accounting • PnL analytics • multi-wallet sub-account tracking
29	● Web3 Auth	Biometric & Keypair Web3 Login Portal	Web3 Authentication	High-security onboarding portal supporting Web3 wallets (Sui Wallet Ethos Slush) and cryptographic keypair authentication.	Obsidian cyber styling • holographic inputs • WebAuthn integration readiness
30	● Design System	Obsidian Cyberpunk Design Tokens	Universal Styling (theme.css)	Unified luxury dark-mode design system with curated HSL colors glassmorphic cards and glowing neon accents.	Primary: #02040a • Emerald: #00ff88 • Cyan: #38bdf8 • Amber: #f59e0b • Rose: #f43f5e
31	● Typography	Space Grotesk & JetBrains Mono	Typography Tokens	Premium Google font pairing combining a geometric grotesque for titles and a monospaced font for telemetry and code.	Variable weights 400-800 • tabular numbers • high legibility for financial data
32	● Backend Server	Node.js & Express.js REST Engine	Server Architecture (server.js)	High-performance backend handling cryptographic blob sharding file retrieval Merkle tree calculation and static asset delivery.	express.json({ limit: 50mb }) • CORS headers • modular REST API routing
33	● Persistence	Physical Shard Storage & Registry	Backend Storage (data/walrus_vault)	Local filesystem persistence engine storing discrete binary shards (.bin) and metadata index (blobs.json).	Binary stream I/O • hash indexing • disk synchronization • atomic writes
34	● Deployment	Vercel Edge Configuration	Cloud Infrastructure (vercel.json)	Cloud deployment configuration routing dynamic API calls and serving high-frequency client bundles with edge caching.	Serverless functions • rewrite rules • CDN static asset caching
35	● Architecture Style	Pure Vanilla High-Performance Architecture	Core Codebase Philosophy	Zero-bloat engineering methodology relying on pure native HTML5 Vanilla JavaScript (ES6+) and CSS3 for instantaneous loads.	Zero heavy framework overhead • 60fps animations • sub-10ms DOM response